AI for IT - SDLC TCS Internal

DURATION: 8-10 HOURS

DEVELOP SOFTWARE USING ARTIFICIAL INTELLIGENCE (AI) TOOLS/AI AGENTS FOR THE FOLLOWING USE STORY

USE STORY TITLE

Strengthen Client Retention and Producer Effectiveness

Challenge setup	Requirement
Teams	1 team of 5 members; 4 teams participate across four separate challenges
Case company	Meridian Brokerage Group, a fictional insurance broker
Anchor business journey	Renewal-to-Retention
Data	Synthetic data only, created by each team
Primary experience	Client growth and retention workspace with evidence-grounded meeting preparation
Executive objective	Growth and Profitability
Executive question	Which client conversation or renewal needs attention, and what evidence should guide the producer?

Evaluation Criteria (Read this, Very Important)

The evaluation focuses on the quality and completeness of each deliverable, the purposeful use of AI tools and AI agents across the Software Development Life Cycle, and the strength of the working software. Every deliverable must be professionally presented, internally consistent and demonstrably complete. Maintain a comprehensive prompt history identifying the tool or model, purpose, key iteration, output accepted or rejected, and human validation performed.

Key Areas of Assessment

- **Comprehensive Application of AI:** Use AI in analysis, requirements, architecture, data modelling, development, testing, documentation and quality review.
- **Impactful Implementation:** Show how AI improves producer preparation, client retention, renewal outcomes and responsible opportunity identification.
- **Deliverable Quality and Completeness:** Provide production-oriented deliverables with professional presentation and traceability.
- **Innovative Approach:** Reward practical innovation in multi-source synthesis, risk detection, coverage insight, next-best conversation and explainable recommendations.
- **Business Relevance:** Connect insights to business outcomes, deadlines, commitments, risk and decisions.
- **Responsible AI:** Use grounded evidence, transparent assumptions, confidence, access controls, human approval and audit trails.
- **Tokenomics and Model Economics:** Measure AI consumption, cost and latency; show fit-for-purpose model selection and optimisation.

Indicative List of Tools That Can Be Used

- **AI App & Workflow Builders:** Lovable AI; Bolt.new; Replit; Vercel v0
- **LLM & Agent Development Platforms:** Google AI Studio (Gemini); OpenAI Playground; Azure AI Studio; Hugging Face Spaces; LangChain; LlamaIndex; CrewAI; AutoGen
- **AI Coding & Code-Quality Agents:** Codeium; Cursor; Continue.dev; Semgrep; SonarQube Community Edition
- **AI for Requirements, Documentation & Analysis:** ChatGPT; Claude; Notion AI; Obsidian with community AI plug-ins
- **AI for UI / UX & Wireframe Design:** Lovable AI; Uizard; Figma; Penpot; Diagrams.net; Whimsical
- **AI for Testing & QA:** Postman AI; Playwright with AI assistance; Selenium; OpenTest
- **AI for Architecture, Data Modeling & Diagrams:** DBdiagram.io; Creately AI; PlantUML
- **AI Agent UI / Demo Frameworks:** Streamlit; Gradio; Chainlit
- **Local / Open-Source Models & Runtimes:** Ollama; LM Studio; Mistral open models; approved hosted inference services

TOOL SELECTION

The list is indicative. Teams must use only event-approved tools, explain selections, and must not upload real customer, employee, policy, payment or credential data.

User Story

Strengthen Client Retention and Producer Effectiveness

Underlying solution: Producer and Client Intelligence Copilot for Meridian Brokerage Group

Meridian's growth depends on retaining client relationships, enabling producers and account teams, and identifying responsible opportunities to deepen those relationships. Yet client, policy, claims, billing, service, activity and document information is distributed across multiple systems and acquired agencies. Producers may spend valuable time assembling context and can enter renewal or stewardship meetings without a complete view of commitments, risk or opportunity.

As a Meridian producer or account executive, I want a concise, evidence-grounded client brief before a renewal or stewardship meeting, so that I can focus the conversation on the client's priorities, address service or renewal risks early, fulfil open commitments and identify appropriate coverage or relationship opportunities without unsupported recommendations.

The experience should improve preparation and decision quality, not replace producer judgement. Every risk or opportunity must show supporting evidence, recency, confidence, assumptions and any contradiction that requires human review.

Expanded Details

- **Client 360 Assembly:** Combine synthetic account, policy, claims, activity, service, billing and document data into a traceable profile.
- **Meeting Preparation:** Generate a one-page brief with relationship summary, upcoming milestones, open commitments, recent changes and suggested discussion points.
- **Renewal Risk Detection:** Identify late exposure information, open claims, service issues, unresolved endorsements, carrier constraints and approaching deadlines.
- **Coverage and Opportunity Signals:** Surface evidence-based coverage gaps, product adjacency and referral opportunities while clearly separating facts from suggestions.
- **Persona Adaptation:** Provide producer, account executive, service leader and regional leader views based on the same evidence.
- **Responsible Recommendations:** Display evidence, confidence, assumptions and required human review; never bind coverage, alter policy data or contact a client autonomously.

Functional Scope

- **Synthetic client data hub:** Accounts, contacts, policies, claims, activities, service cases and documents.
- **Client intelligence model:** Relationship timeline, renewal milestones, issues, commitments and opportunity signals.
- **Brief generator:** Executive summary, agenda, talking points, risks and follow-ups.
- **Evidence panel:** Source record, recency, confidence and contradiction handling.
- **Conversational copilot:** Grounded questions, drill-down and persona-sensitive answers.
- **Controlled next actions:** Draft follow-up, create review task or prepare referral subject to approval.
- **Feedback and quality:** User corrections, accepted/rejected suggestions and accuracy monitoring.
- **Audit and economics:** Prompt/model trace, access logging, token usage, cost and latency.

Mandatory Scenarios

- Material open claim omitted from initial summary
- Conflicting renewal dates across two systems
- Apparent cross-sell opportunity lacks sufficient evidence
- Prompt attempts to expose another client's restricted information

LIVE SCENARIO

Judges may inject an additional scenario during the demonstration. The solution must accept the new scenario without code-level rework to the core experience.

Deliverables

- **1. Functional Requirements Document:** User story, expanded requirements, functional and non-functional requirements, assumptions, acceptance criteria, personas and out-of-scope items.
- **2. Application Architecture and Architecture Document:** Context, components, integrations, AI/agent flow, security, deployment, observability and scale considerations.
- **3. Database Design / Data Models:** Use an RDBMS. Provide logical and physical models, ER diagram, data dictionary and repeatable synthetic-data design. Do not use JSON as the primary input store.
- **4. UI / UX / Wireframe Design:** Primary dashboard or cockpit, drill-downs, conversational interface, approvals and outcome views.
- **5. Software Code & Working Software:** Runnable end-to-end prototype with repository, configuration, setup guide and repeatable synthetic-data generator.
- **6. Functional and Non-Functional Test Scripts:** CSV or Excel covering positive, negative, security, performance, grounding, low-confidence, token/cost and live-scenario tests.
- **7. Traceability Matrix:** CSV or Excel linking requirements, architecture/components, code/modules, tests and demonstration evidence.
- **8. End User Manual:** Screens, persona guidance, usage steps, approval flow, assumptions, limitations and troubleshooting.
- **9. AI Agent - Producer and Client Intelligence Agent:** Build using an AI framework. The agent accepts client, policy, claims, service and activity context and produces a concise brief, risk flags, confidence, evidence links and controlled next-best conversation suggestions.
- **10. Prompt and AI Usage Register:** Record prompts used across requirements, design, development, testing, documentation and review, including human validation.
- **11. Tokenomics and Model Economics Report:** Show usage, cost, latency, model rationale, optimisation controls and scaled-volume projection.
- **12. Demo Package:** Provide architecture summary, assumptions, known limitations, an 8-minute demonstration script and a 2-minute business-value statement.

PRODUCER AND CLIENT INTELLIGENCE AGENT

The agent is mandatory and must be integrated with the principal application experience. Streamlit, Gradio or Chainlit may support a separate agent demonstration but must not be the only user interface.

Instructions

- **Selection of Tools:** Evaluate and select suitable AI tools or frameworks for each SDLC phase and explain the rationale.
- **Quality Standards:** Follow coding standards, secure-design principles, consistent design patterns, code review and automated testing.
- **Integration:** Ensure seamless integration across data, business logic, AI agent, user experience, controlled action flow and audit.
- **Testing:** Cover incomplete data, contradictory signals, security, performance, low confidence, grounding, cost and live scenarios.
- **Documentation:** Maintain current requirements, architecture, data model, prompts, tests, traceability, user guidance and limitations.
- **User Experience:** Prioritise a polished primary application. Streamlit or Gradio may be used for a separate agent demonstration, not as the only UI.
- **Synthetic Data:** Use only fictional names and synthetic business, policy, payment, agency, carrier, employee and credential data.
- **Responsible AI:** Ground conclusions in evidence, show confidence and assumptions, apply least-privilege access, require human approval for material actions and maintain an audit trail.
- **Prompt History:** Retain complete prompt and revision history where event policy permits.
- **Tokenomics:** Measure usage by major workflow and demonstrate cost, latency and quality optimisation.

Evaluation Rubric

Criterion	Points
Business relevance	15
Brief quality and usefulness	15
Risk detection	15
Grounding and evidence	10
End-to-end engineering	10
AI reasoning quality	10
Trust, privacy and Responsible AI	10
User experience	10
Tokenomics and model economics	5

Demonstration Guidance

Open with the business outcome and executive question. Show the decision that must be made, the evidence and exceptions that matter, the accountable action and approval path, and the measurable business outcome. Explain the AI, data and technical design only after the business story is clear.

FINAL EXPECTATION
Build a client-growth and retention experience, not a generic chatbot. Demonstrate faster preparation, stronger client conversations, early renewal-risk intervention, traceable opportunity signals and human-controlled recommendations using only synthetic data.